CORE PURE / DIFFERENTIAL EQUATIONS

Modelling with 2nd Order Differential Equations

Mark scheme 12 questions · 162 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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	Total	**162**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

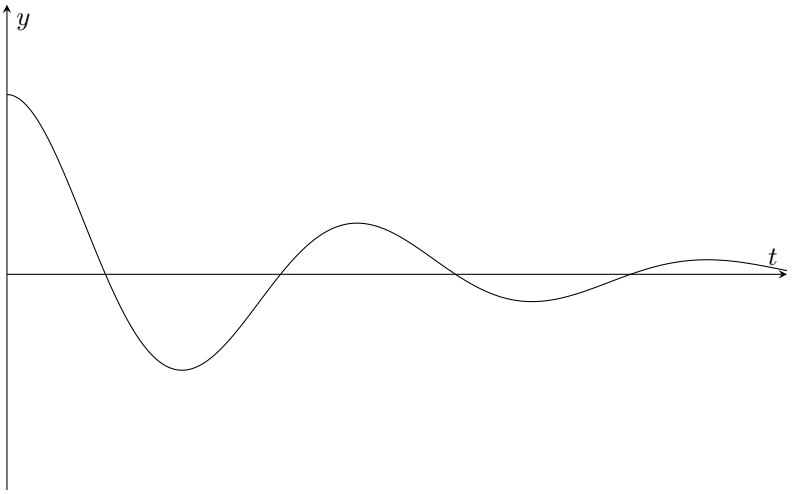
- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)(i)	$p = 0$	**B1**	
		[1]	
1(a)(ii)	Real suspension is damped, so the amplitude decreases; simple harmonic motion keeps a constant amplitude	**E1**	States that damping (shock absorbers, friction, energy loss) makes oscillations die away, unlike simple harmonic motion's constant amplitude. Either idea alone, clearly expressed, is enough.
		[1]	
1(b)	$4\lambda^2 + p\lambda + 9 = 0$ $p^2 - 4(4)(9) \geq 0 \Rightarrow p \ldots$	**M1**	Uses the discriminant of the auxiliary equation with the condition for real roots. Condone $>$ for $\geq$ and a numerical slip.
	$p \geq 12$	**A1**	Must include $p = 12$ and no negative values; condone $p \leq -12$ also stated if $p \geq 12$ is then given as the answer.
		[2]	
1(c)			
	Shape	**B1**	Oscillation about the t-axis with decreasing amplitude, starting at positive y with zero gradient and crossing the axis.
		[1]	
	Total for Question 1	**5**	

Question	Scheme	Marks	Notes
2(a)	$\lambda^2 + \lambda - 6 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by factorising with product of constants -6, the formula or completing the square; calculator allowed.
	$\lambda = -3,\ 2$ $(E =)\ Ae^{-3t} + Be^{2t}$	**A1**	
	$-6k = -12 \Rightarrow k = 2$	**B1**	
	E tends to a finite limit $\Rightarrow B = 0$	**B1**	$B = 0$ stated or used, the coefficient of e^{2t} removed.
	$\frac{\mathrm{d}E}{\mathrm{d}t} = -3Ae^{-3t} \Rightarrow -3A = -9 \Rightarrow A = \ldots$ $E =$ "3"$e^{-3\ln 2}$ + "2"	**M1**	Differentiates their solution of the form $Ae^{-3t} + c$, uses $\frac{\mathrm{d}E}{\mathrm{d}t} = -9$ at $t = 0$ to find A, and substitutes $t = \ln 2$. Condone sign slips.
	$E = \frac{19}{8}$	**A1**	$\frac{19}{8}$ oe, such as 2.375.
		[6]	
2(b)	$E \to 2$	**B1ft**	ft their constant particular integral, for a solution whose remaining exponential term decays.
		[1]	
	Total for Question 2	**7**	

Question	Scheme	Marks	Notes
3(a)	$3k(-0.10) + 2k(0) + 5(0.18) = 0 \Rightarrow k = \ldots$	**M1**	Substitutes all three initial values into the differential equation. Condone a sign slip.
	$k = 3$	**A1**	
		[2]	
3(b)	$9\lambda^2 + 6\lambda + 5 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation with their positive k and solves it by any means; calculator allowed.
	$\lambda = -\frac{1}{3} \pm \frac{2}{3}i$ $x = e^{-t/3}\left(A\cos\frac{2t}{3} + B\sin\frac{2t}{3}\right)$	**A1**	oe, such as $Re^{-t/3}\cos\left(\frac{2t}{3} - \alpha\right)$.
	$t = 0,\ x = 0.18 \Rightarrow A = 0.18$	**M1**	Uses $x = 0.18$ at $t = 0$ in their general solution with two arbitrary constants to find one constant.
	$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{1}{3}e^{-t/3}\left(A\cos\frac{2t}{3} + B\sin\frac{2t}{3}\right)$ $\quad + e^{-t/3}\left(-\frac{2A}{3}\sin\frac{2t}{3} + \frac{2B}{3}\cos\frac{2t}{3}\right)$ $t = 0:\ 0 = -\frac{1}{3}A + \frac{2}{3}B \Rightarrow B = \ldots$	**M1**	Differentiates their solution of the form $e^{pt}(A\cos qt + B\sin qt)$ by the product rule, sets $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ at $t = 0$ and solves. Condone coefficient slips.
	$x = e^{-t/3}\left(0.18\cos\frac{2t}{3} + 0.09\sin\frac{2t}{3}\right)$	**A1**	oe, exact or decimal coefficients; needs $x = \ldots$
		[5]	
3(c)	$1.2 + e^{-1.5}$ ("0.18" $\cos 3 +$ "0.09" $\sin 3$)	**M1**	Substitutes $t = 4.5$ into their particular solution and adds 1.2.
	$= 1.16$ m	**A1**	awrt 1.16
		[2]	
3(d)	Spring and damping forces assumed proportional to displacement and velocity	**E1**	States any valid limitation, such as forces not exactly linear, spring exceeding its elastic limit, or the track's length limiting the motion. Not for a bare "unrealistic".
		[1]	
	Total for Question 3	**10**	

Question	Scheme	Marks	Notes
4(a)(i)	$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -9mx$ $\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 9x = 0$	**B1**	Needs $F = ma$ with the force as $-9mx$, leading to the printed equation.
		[1]	
4(a)(ii)	Simple harmonic motion	**B1**	Accept SHM.
		[1]	
4(a)(iii)	$T = \frac{2\pi}{3}$ (s)	**B1**	Or awrt 2.09.
		[1]	
4(a)(iv)	$\lambda^2 + 9 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it, or quotes the solution form for $\ddot{x} = -\omega^2 x$ with $\omega = 3$. Implied by a correct general solution.
	$\lambda = \pm 3i$ $x = A\cos 3t + B\sin 3t$	**A1**	Needs $x =$; any letters for the constants.
	$t = 0,\ x = 1 \Rightarrow A = \ldots\ [= 1]$	**M1**	Substitutes $t = 0$, $x = 1$ into their general solution with two arbitrary constants.
	$\frac{\mathrm{d}x}{\mathrm{d}t} = -3A\sin 3t + 3B\cos 3t$ $t = 0,\ \frac{\mathrm{d}x}{\mathrm{d}t} = -6 \Rightarrow B = \ldots\ [= -2]$	**M1**	Differentiates their general solution and substitutes $t = 0$ with $\frac{\mathrm{d}x}{\mathrm{d}t} = \pm 6$.
	$x = \cos 3t - 2\sin 3t$	**A1**	
		[5]	
4(a)(v)	Amplitude $= \sqrt{\text{“1”}^2 + \text{“}(-2)\text{”}^2}$	**M1**	Uses $\sqrt{A^2 + B^2}$ with their A and B, or finds R in $R\cos(3t + \alpha)$ oe.
	$= \sqrt{5}$ (m)	**A1**	Or awrt 2.24.
		[2]	
4(b)(i)	$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -6m\frac{\mathrm{d}x}{\mathrm{d}t} - 9mx$ $\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6\frac{\mathrm{d}x}{\mathrm{d}t} + 9x = 0$	**B1**	Needs $F = ma$ with both forces signed correctly, $-6m\frac{\mathrm{d}x}{\mathrm{d}t}$ and $-9mx$, leading to the printed equation.
		[1]	
4(b)(ii)	$\lambda^2 + 6\lambda + 9 = 0$: $6^2 - 4 \times 9 = 0$ Repeated root, so critically damped	**B1**	Critically damped, with a reason: a repeated root, or discriminant shown to be zero.
		[1]	

Question 4 continued

Question	Scheme	Marks	Notes
4(b)(iii)	$\lambda^2 + 6\lambda + 9 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by any means; may be seen in (b)(ii).
	$\lambda = -3$ (repeated)	**A1**	
	$x = (A + Bt)e^{-3t}$	**A1**	Needs $x =$; any letters for the constants.
		[3]	
4(c)(i)	$(x =)\ a\cos 4t + b\sin 4t$	**B1**	
	$-16a\cos 4t - 16b\sin 4t + 6(-4a\sin 4t + 4b\cos 4t)$ $+ 9(a\cos 4t + b\sin 4t) = 25\cos 4t$	**M1**	Differentiates their trial function twice, substitutes into the differential equation and equates coefficients of $\cos 4t$ and $\sin 4t$. Condone sign and coefficient slips.
	$-7a + 24b = 25, \quad -24a - 7b = 0$	**A1**	Both equations correct, oe.
	$\Rightarrow a = \ldots,\ b = \ldots$	**dM1**	Solves their pair of simultaneous equations for a and b by any means.
	$a = -\frac{7}{25},\ b = \frac{24}{25}$	**A1**	
	$x = (A + Bt)e^{-3t}$ "$-\frac{7}{25}\cos 4t + \frac{24}{25}\sin 4t$" $t = 0,\ x = 1 \Rightarrow A = \ldots\ [= \frac{32}{25}]$	**M1**	Substitutes $t = 0$, $x = 1$ into a general solution of the form $(A + Bt)e^{-3t}+$ their particular integral.
	$\frac{\mathrm{d}x}{\mathrm{d}t} = (B - 3A - 3Bt)e^{-3t}$ $+ \frac{28}{25}\sin 4t + \frac{96}{25}\cos 4t$ $t = 0,\ \frac{\mathrm{d}x}{\mathrm{d}t} = -6 \Rightarrow B = \ldots\ [= -6]$	**M1**	Differentiates their general solution, using the product rule on Bte^{-3t}, and substitutes $t = 0$ with $\frac{\mathrm{d}x}{\mathrm{d}t} = \pm 6$.
	$x = \left(\frac{32}{25} - 6t\right)e^{-3t}$ $-\frac{7}{25}\cos 4t + \frac{24}{25}\sin 4t$	**A1**	oe; needs $x =$.
		[8]	
4(c)(ii)	$\left(\frac{32}{25} - 6t\right)e^{-3t} \to 0,\ \omega = 4$ $T = \frac{2\pi}{4}$	**M1**	States the exponential terms tend to 0 and uses $T = \frac{2\pi}{\omega}$ with ω from their particular integral.
	$= \frac{\pi}{2} \approx 1.57$, consistent with 1.6	**A1**	$\frac{\pi}{2}$ or awrt 1.57, with a conclusion that this is consistent with 1.6.
		[2]	
	Total for Question 4	**25**	

Question	Scheme	Marks	Notes
5(a)	$\lambda^2 + 2\lambda + 10 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by any method; calculator allowed.
	$\lambda = -1 \pm 3i$ $q = e^{-t}(A\cos 3t + B\sin 3t)$	**A1**	
	$q = a\cos 3t + b\sin 3t$	**B1**	
	$-9a\cos 3t - 9b\sin 3t + 2(-3a\sin 3t + 3b\cos 3t)$ $\quad + 10(a\cos 3t + b\sin 3t) = 37\cos 3t$	**M1**	Differentiates a trial function of the form $a\cos 3t + b\sin 3t$ twice, substitutes and equates coefficients of $\cos 3t$ and $\sin 3t$. Condone sign and coefficient slips.
	$a + 6b = 37, \quad b - 6a = 0$	**A1**	
	$a = 1, \quad b = 6$	**A1**	
	$q = e^{-t}(A\cos 3t + B\sin 3t) + \cos 3t + 6\sin 3t$	**A1ft**	ft their complementary function, with two arbitrary constants, plus their particular integral with numerical coefficients.
	$1 = A + 1 \Rightarrow A = \ldots$	**M1**	Substitutes $t = 0$, $q = 1$ into their general solution with two arbitrary constants.
	$\frac{\mathrm{d}q}{\mathrm{d}t} = -e^{-t}(A\cos 3t + B\sin 3t)$ $\quad + e^{-t}(-3A\sin 3t + 3B\cos 3t)$ $\quad - 3\sin 3t + 18\cos 3t$ $15 = 3B - A + 18 \Rightarrow B = \ldots$	**M1**	Differentiates their general solution, using the product rule on the exponential term, and substitutes $t = 0$, $\frac{\mathrm{d}q}{\mathrm{d}t} = 15$. Condone sign slips.
	$q = \cos 3t + 6\sin 3t - e^{-t}\sin 3t$	**A1**	
		[10]	
5(b)	$q \approx \cos 3t + 6\sin 3t$	**B1**	No justification for discarding $e^{-t}\sin 3t$ is needed.
	$R\cos\alpha = 1, \quad R\sin\alpha = 6$ $\Rightarrow R = \ldots, \quad \tan\alpha = \ldots$	**M1**	Compares coefficients for their $q \approx p\cos 3t + r\sin 3t$ and finds R or α. Condone sign slips or $\tan\alpha = \frac{1}{6}$.
	$R = \sqrt{37}$	**A1**	Or awrt 6.08.
	$q \approx \sqrt{37}\cos(3t - \arctan 6)$	**A1**	$\alpha = \arctan 6$ or awrt 1.41, in the stated form $R\cos(3t - \alpha)$. Values of R and α alone, or $80.5°$, do not score.
		[4]	
	Total for Question 5	**14**	

Question	Scheme	Marks	Notes
6(a)	$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -c\frac{\mathrm{d}y}{\mathrm{d}t} - ky$ $\Rightarrow m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + c\frac{\mathrm{d}y}{\mathrm{d}t} + ky = 0$	**M1 A1**	**M1** Applies Newton's second law with a spring force proportional to y and a resistance proportional to $\frac{\mathrm{d}y}{\mathrm{d}t}$, weight absent or cancelled by the equilibrium tension. Condone sign slips. **A1** Printed equation, with both forces correctly signed against displacement and motion.
	$m\lambda^2 + c\lambda + k = 0$ $\Rightarrow \lambda = \frac{-c\pm\sqrt{c^2-4mk}}{2m}$	**M1**	Forms the auxiliary equation and solves it by the quadratic formula or completing the square. Condone sign slips.
	$\lambda = -\frac{c}{2m} \pm i\frac{\sqrt{4km-c^2}}{2m}$	**A1**	Needs $\sqrt{c^2-4km}$ written as $i\sqrt{4km-c^2}$, oe.
	$y = e^{-\frac{ct}{2m}}\left(P\cos\left(\frac{\sqrt{4km-c^2}}{2m}t\right)\right.$ $\left. + Q\sin\left(\frac{\sqrt{4km-c^2}}{2m}t\right)\right)$	**A1 AG**	Printed solution from correct complex roots via $e^{\alpha t}(P\cos\beta t + Q\sin\beta t)$, with no errors.
		[5]	
6(b)(i)	$y = e^{-3t}(P\cos 4t + Q\sin 4t)$	**B1**	
	$t = 0,\ y = 0.12 \Rightarrow P = \ldots$	**M1**	Substitutes $t = 0$, $y = 0.12$ into their solution of the form $e^{-at}(P\cos bt + Q\sin bt)$.
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -3e^{-3t}(P\cos 4t + Q\sin 4t)$ $+ e^{-3t}(-4P\sin 4t + 4Q\cos 4t)$ $t = 0 : -3P + 4Q = -0.12 \Rightarrow Q = \ldots$	**M1**	Differentiates their solution by the product rule and substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = -0.12$. Condone sign slips.
	$P = 0.12,\ Q = 0.06$	**A1**	
		[4]	
6(b)(ii)	$e^{-3t}(0.12\cos 4t + 0.06\sin 4t) = 0$ $\Rightarrow 0.12\cos 4t + 0.06\sin 4t = 0$	**M1**	Sets their $y = 0$ and removes the non-zero exponential factor.
	$\tan 4t = -2$	**A1**	oe, including a correct harmonic form equated to zero.
	$4t = \pi - \arctan 2\ [= 2.034\ldots]$	**M1 A1**	**M1** Finds the smallest positive $4t$ satisfying their equation, e.g. adds π to a negative principal value. **A1** oe, or awrt 2.03.
	$t = \frac{\pi - \arctan 2}{4} \approx 0.509$	**A1**	Exact form or awrt 0.509.
		[5]	

Question 6 continued

Question	Scheme	Marks	Notes
6(c)	$5\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 125y = 0$ $\Rightarrow 5\lambda^2 + 125 = 0 \Rightarrow \lambda = \ldots$	**M1**	Sets $c = 0$ and solves the auxiliary equation by any method.
	$\lambda = \pm 5i$ $y = A\cos 5t + B\sin 5t$	**A1**	
	$t = 0,\ y = 0.12 \Rightarrow A = \ldots$	**M1**	Substitutes $t = 0$, $y = 0.12$ into their solution of the form $A\cos\omega t + B\sin\omega t$.
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -5A\sin 5t + 5B\cos 5t$ $t = 0 : 5B = -0.12 \Rightarrow B = \ldots$	**M1**	Differentiates their solution and substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = -0.12$. Condone sign slips.
	$y = 0.12\cos 5t - 0.024\sin 5t$	**A1**	y in terms of t, oe such as a correct harmonic form, constants to 2 s.f.; A0 for A and B stated alone.
		[5]	
	Total for Question 6	**19**	

Question	Scheme	Marks	Notes
7(a)	$4\lambda^2 + 4\lambda + 5 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it: correct formula with their values, completing the square, or calculator.
	$\lambda = -\frac{1}{2} \pm i$ $(m =)\ e^{-t/2}(A\cos t + B\sin t)$	**A1**	oe, such as $Re^{-t/2}\cos(t+\alpha)$.
	$5k = 20 \Rightarrow k = 4$	**B1**	
	$m = 4 + e^{-t/2}(A\cos t + B\sin t)$	**A1ft**	ft their complementary function plus their constant particular integral; must be $m = \ldots$ in terms of t.
		[4]	
7(b)	$t = 0,\ m = 0:\ 0 = 4 + A \Rightarrow A = \ldots\ [= -4]$	**M1**	Substitutes $t = 0$, $m = 0$ into their general solution to find a constant.
	$\frac{\mathrm{d}m}{\mathrm{d}t} = -\frac{1}{2}e^{-t/2}(A\cos t + B\sin t)$ $\quad + e^{-t/2}(-A\sin t + B\cos t)$ $5 = -\frac{1}{2}A + B \Rightarrow B = \ldots$	**M1**	Differentiates their general solution by the product rule, both terms present, and substitutes $t = 0$, $\frac{\mathrm{d}m}{\mathrm{d}t} = 5$. Condone sign slips.
	$m = 4 + e^{-t/2}(-4\cos t + 3\sin t)$	**A1**	
	$\frac{\mathrm{d}m}{\mathrm{d}t} = e^{-t/2}\left(\frac{5}{2}\sin t + 5\cos t\right) = 0$	**M1**	Differentiates their particular solution by the product rule and sets the derivative to 0. Need not be simplified.
	$\tan t = -2$	**A1**	oe
	$t = \pi - \arctan 2\ [\approx 2.03]$	**A1**	oe, or awrt 2.03. No justification that this gives the greatest value is required.
	$m = 4 + e^{-\text{“2.03”}/2}\left(-4\cos\text{“2.03”} + 3\sin\text{“2.03”}\right)$ $\quad = \ldots$	**dM1**	Substitutes their first positive root of $\frac{\mathrm{d}m}{\mathrm{d}t} = 0$ into their particular solution.
	$m_{\max} \approx 5.62\ (\text{kg})$	**A1**	awrt 5.62, or exact $4 + 2\sqrt{5}\,e^{-(\pi - \arctan 2)/2}$.
		[8]	
7(c)	$m(8) = 4 + e^{-4}(-4\cos 8 + 3\sin 8) = \ldots$	**M1**	Substitutes $t = 8$ into their particular solution from (b), in radians, and evaluates.

Question 7 continued

Question	Scheme	Marks	Notes
	$m(8) \approx 4.065 > 4.05$, so no label is printed	**A1**	awrt 4.07 (or 4.065), compared with 4.05, and concludes no label is printed.
		[2]	
	Total for Question 7	**14**	

Question	Scheme	Marks	Notes
8(a)	$\lambda^2 + 5\lambda + 6 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by any method; calculator allowed.
	$(\theta =) Ae^{-2t} + Be^{-3t}$	**A1**	
	$\theta = ae^{-4t}$	**B1**	
	$16ae^{-4t} - 20ae^{-4t} + 6ae^{-4t} = 4e^{-4t}$ $\Rightarrow a = \ldots$	**M1**	Differentiates their particular integral of the form ae^{-4t} twice and substitutes into the equation. Condone coefficient slips.
	$a = 2$	**A1**	
	$\theta = Ae^{-2t} + Be^{-3t} + 2e^{-4t}$	**A1ft**	ft their complementary function and their a; needs $\theta =$ and two arbitrary constants.
	$t = 0,\ \theta = 0 \Rightarrow A + B + 2 = 0$	**M1**	Substitutes $t = 0$, $\theta = 0$ into their general solution with two arbitrary constants.
	$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -2Ae^{-2t} - 3Be^{-3t} - 8e^{-4t}$ $-2A - 3B - 8 = -1 \Rightarrow A = \ldots,\ B = \ldots$	**M1**	Differentiates their general solution, uses $\frac{\mathrm{d}\theta}{\mathrm{d}t} = \pm 1$ at $t = 0$ and solves both equations for A and B.
	$\theta = e^{-2t} - 3e^{-3t} + 2e^{-4t}$	**A1**	
		[9]	
8(b)	$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -2e^{-2t} + 9e^{-3t} - 8e^{-4t} = 0$	**M1**	Differentiates their θ, with at least one term $ke^{pt} \rightarrow kpe^{pt}$, and sets the derivative equal to zero.
	$2e^{2t} - 9e^{t} + 8 = 0$	**A1**	Correct quadratic in e^t, any multiple or equivalent.
	$e^t = \frac{9+\sqrt{17}}{4} \Rightarrow t = \ldots\ [= 1.188]$ $\theta = e^{-2t} - 3e^{-3t} + 2e^{-4t} = \ldots$	**dM1**	Solves for e^t, takes logarithms and substitutes the root giving positive θ into their θ; trying both roots also scores. No justification of the choice needed.
	0.025 (rad)	**A1**	awrt 0.025
		[4]	
8(c)	$e^{-4t}(e^t - 1)(e^t - 2) = 0 \Rightarrow t = \ldots$	**M1**	Sets their $\theta = 0$ and solves for a positive t, e.g. as a quadratic in e^t; calculator allowed.

Question 8 continued

Question	Scheme	Marks	Notes
	$t = \ln 2 \approx 0.693$, close to 0.69, so supports the model	**A1**	$\ln 2$ or awrt 0.693, with a conclusion that the model is supported because it agrees with 0.69 to 2 d.p. oe.
		[2]	
	Total for Question 8	**15**	

Question	Scheme	Marks	Notes
9(a)	$\lambda^2 + 2\lambda + 10 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the three-term auxiliary equation and solves it by formula, completing the square or calculator.
	$\lambda = -1 \pm 3i$ $(y =)\ e^{-t}(A\cos 3t + B\sin 3t)$	**A1**	oe, in t.
	$y = C\cos t + D\sin t$	**B1**	
	$(-C\cos t - D\sin t) + 2(-C\sin t + D\cos t)$ $+ 10(C\cos t + D\sin t) = 20\cos t + 5\sin t$	**M1**	Differentiates their $C\cos t + D\sin t$ twice, substitutes into the equation and equates coefficients of $\cos t$ and $\sin t$. Condone sign slips.
	$9C + 2D = 20, \quad 9D - 2C = 5$	**A1**	Both equations, oe.
	$C = 2, \quad D = 1$	**A1**	
	$y = e^{-t}(A\cos 3t + B\sin 3t) + 2\cos t + \sin t$	**A1ft**	ft their complementary function of the form $e^{pt}(A\cos qt + B\sin qt)$ plus their particular integral; needs $y =$ and in t.
		[7]	
9(b)	$3 = A + 2 \Rightarrow A = \ldots$	**M1**	Substitutes $t = 0$, $y = 3$ into their general solution of the form complementary function plus particular integral to find A.
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -e^{-t}(A\cos 3t + B\sin 3t)$ $+ e^{-t}(-3A\sin 3t + 3B\cos 3t) - 2\sin t + \cos t$ $3 = -A + 3B + 1 \Rightarrow B = \ldots$	**M1**	Differentiates their general solution, using the product rule on the exponential term, and substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 3$. Condone sign slips.
	$y = e^{-t}(\cos 3t + \sin 3t) + 2\cos t + \sin t$	**A1**	
		[3]	
9(c)	$e^{-t}(2\cos 3t - 4\sin 3t) + \cos t - 2\sin t = 0$ $\Rightarrow t = \ldots$	**M1**	Sets the derivative of their particular solution equal to 0 and finds its smallest positive root by calculator or iteration.
	$t = 0.2068$	**A1**	awrt 0.207
	$a = e^{-0.2068}(\cos 0.6204 + \sin 0.6204)$ $+ 2\cos 0.2068 + \sin 0.2068$	**dM1**	Substitutes their t into their particular solution from (b).

Question 9 continued

Question	Scheme	Marks	Notes
	$a = 3.30$	**A1**	awrt 3.30
		[4]	
	Total for Question 9	**14**	

Question	Scheme	Marks	Notes
10(a)(i)	$x = t(a\cos 4t + b\sin 4t)$ $\frac{\mathrm{d}x}{\mathrm{d}t} = a\cos 4t + b\sin 4t$ $\quad + t(-4a\sin 4t + 4b\cos 4t)$ $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \ldots$	**M1**	Differentiates a trial of the form $t(a\cos 4t + b\sin 4t)$, or the given x, twice using the product rule. Condone sign and coefficient slips.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -8a\sin 4t + 8b\cos 4t$ $\quad - 16t(a\cos 4t + b\sin 4t)$	**A1**	Any form; or $3\sin 4t + 6t\cos 4t$ from the given x.
	$-8a\sin 4t + 8b\cos 4t = 3\sin 4t$ $\Rightarrow a = \ldots,\ b = \ldots$	**M1**	Substitutes into the equation and compares coefficients of $\sin 4t$ and $\cos 4t$; or, for the given x, substitutes and simplifies the LHS.
	$a = -\frac{3}{8},\ b = 0 \Rightarrow x = -\frac{3}{8}t\cos 4t$	**A1 AG**	From a correct substituted equation without error. Verification route: LHS shown to equal $3\sin 4t$, with LHS = RHS stated.
		[4]	
10(a)(ii)	$\lambda^2 + 16 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by any method.
	$\lambda = \pm 4i$	**A1**	
	$x = A\cos 4t + B\sin 4t$	**A1**	oe, such as $R\cos(4t + \alpha)$.
	$x = A\cos 4t + B\sin 4t - \frac{3}{8}t\cos 4t$	**A1ft**	ft their complementary function with two arbitrary constants, plus $-\frac{3}{8}t\cos 4t$; must be $x =$ a function of t.
		[4]	
10(b)	$t = 0,\ x = 0.20 \Rightarrow A = 0.20$	**M1**	Substitutes $t = 0$, $x = 0.20$ into their general solution with two arbitrary constants.
	$\frac{\mathrm{d}x}{\mathrm{d}t} = -4A\sin 4t + 4B\cos 4t$ $\quad - \frac{3}{8}\cos 4t + \frac{3}{2}t\sin 4t$ $-0.50 = 4B - \frac{3}{8} \Rightarrow B = \ldots$	**M1**	Differentiates their general solution, using the product rule on the $t\cos 4t$ term, and substitutes $t = 0$, $\frac{\mathrm{d}x}{\mathrm{d}t} = \pm 0.5$.
	$x = 0.20\cos 4t - \frac{1}{32}\sin 4t - \frac{3}{8}t\cos 4t$	**A1**	Needs $\frac{\mathrm{d}x}{\mathrm{d}t} = -0.5$ used; $B = -0.03125$ accepted.
	$\beta = 0.20\cos 16 - \frac{1}{32}\sin 16 - \frac{3}{2}\cos 16$ $\quad = 1.25$	**A1**	awrt 1.25
		[4]	

Question 10 continued

Question	Scheme	Marks	Notes
10(c)	Not suitable: model gives 1.25, far more than 0.46	**E1**	States the model is not suitable, comparing 1.25 with 0.46: e.g. overestimates by about 0.8 m, or by more than double. A0 for a conclusion without the comparison.
		[1]	
10(d)	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 16x = 0$	**B1**	oe, such as $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -16x$.
		[1]	
	Total for Question 10	**14**	

Question	Scheme	Marks	Notes
11(a)(i)	$\frac{\mathrm{d}x}{\mathrm{d}t} > 0$, resistance opposes motion $\Rightarrow -2\left\lvert\frac{\mathrm{d}x}{\mathrm{d}t}\right\rvert = -2\frac{\mathrm{d}x}{\mathrm{d}t}$, so term negative	**E1**	States that the velocity is positive (or the trolley moves away from A) and the resistance opposes motion, so the term is negative.
		[1]	
11(a)(ii)	$4(0+1) - 1 - 2(1) = 2\frac{\mathrm{d}^2x}{\mathrm{d}t^2} \Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \ldots$	**M1**	Substitutes $t = 0$, $x = 1$ and $\frac{\mathrm{d}x}{\mathrm{d}t} = 1$ into the equation and solves for the second derivative. Condone one arithmetic slip.
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = \frac{1}{2}$ $(\mathrm{m\,s^{-2}})$	**A1**	
		[2]	
11(b)(i)	$g(0) = 1,\ g'(0) = 1 \Rightarrow g(t) = 1 + t$	**B1**	Needs $g(0) = 1$ and $g'(0) = 1$ used in the Maclaurin form.
		[1]	
11(b)(ii)	$\frac{t^2}{2!}g''(0) = \frac{t^2}{2} \times \frac{1}{2} = \frac{1}{4}t^2$	**B1**	Needs $\frac{g''(0)}{2!}t^2$ with $g''(0) = \frac{1}{2}$ shown.
		[1]	
11(b)(iii)	$4 - \frac{\mathrm{d}x}{\mathrm{d}t} - 2\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 2\frac{\mathrm{d}^3x}{\mathrm{d}t^3}$	**M1 A1**	**M1** Differentiates every term with respect to t, raising each derivative's order by one. Condone sign or coefficient slips. **A1** Correct equation, in x or g.
	$4 - 1 - 2\left(\frac{1}{2}\right) = 2g'''(0) \Rightarrow g'''(0) = \ldots$	**M1**	Substitutes $g'(0) = 1$ and their $g''(0)$ into their differentiated equation, which contains a third derivative, and solves for $g'''(0)$.
	$g'''(0) = 1 \Rightarrow \frac{t^3}{3!} \times 1 = \frac{1}{6}t^3$	**A1 AG**	Correct substituted equation leading without error to $g'''(0) = 1$, then $\frac{g'''(0)}{3!}t^3$ used to reach the printed term.
		[4]	
11(c)(i)	$1 + 0.4 + \frac{1}{4}(0.4)^2 + \frac{1}{6}(0.4)^3 = \ldots$	**M1**	Substitutes $t = 0.4$ into $1 + t + \frac{1}{4}t^2 + \frac{1}{6}t^3$.
	≈ 1.451, close to 1.45, so model suitable	**A1**	awrt 1.451 (or 1.45), compared with 1.45, concludes the model is suitable.
		[2]	

Question 11 continued

Question	Scheme	Marks	Notes
11(c)(ii)	Omitted terms in $t^4, t^5, \ldots$ large at $t = 8$, so cubic inaccurate	**E1**	States that the omitted higher-power terms are large at $t = 8$, or that $t = 8$ is not small. Not for claiming the series is invalid there.
		[1]	
	Total for Question 11	**12**	

Question	Scheme	Marks	Notes
12(a)	$(x =)\ C\cos\lambda t + D\sin\lambda t$	**B1**	oe, such as $R\cos(\lambda t + \alpha)$, with two arbitrary constants.
		[1]	
12(b)	$x(0) = C = -\frac{a}{2},\ C^2 + D^2 = a^2$ $v(0) = D\lambda > 0 \Rightarrow D = \ldots$	**M1**	Uses $x = -\frac{a}{2}$ at $t = 0$ and that the amplitude is a to find values for both constants. Condone the wrong sign of D.
	$x = -\frac{a}{2}\cos\lambda t + \frac{\sqrt{3}}{2}a\sin\lambda t$ $= a\cos\left(\lambda t - \frac{2\pi}{3}\right)$	**A1**	oe, such as $a\sin\left(\lambda t - \frac{\pi}{6}\right)$.
	$v = \frac{a\lambda}{2}\sin\lambda t + \frac{\sqrt{3}}{2}a\lambda\cos\lambda t$ $= -a\lambda\sin\left(\lambda t - \frac{2\pi}{3}\right)$	**A1ft**	ft their x of the form $C\cos\lambda t + D\sin\lambda t$ or $R\cos(\lambda t + \alpha)$ with values found; correct derivative, oe.
		[3]	
12(c)	$v^2 = a^2\lambda^2\sin^2\left(\lambda t - \frac{2\pi}{3}\right)$ $= \lambda^2\left(a^2 - a^2\cos^2\left(\lambda t - \frac{2\pi}{3}\right)\right)$ $= \lambda^2(a^2 - x^2)$	**B1**	Correct v^2 from (b) and $\sin^2 + \cos^2 = 1$ leading without error to the printed result; working in the expanded form accepted.
		[1]	
12(d)	$r_m = \frac{2}{3a}\int_{-a/2}^{a}\frac{1}{\lambda\sqrt{a^2-x^2}}\,\mathrm{d}x$	**B1**	Divisor $\frac{3a}{2}$, limits $-\frac{a}{2}$ and a, and $r = \frac{1}{\lambda\sqrt{a^2-x^2}}$.
	$= \frac{2}{3a\lambda}\left[\arcsin\frac{x}{a}\right]_{-a/2}^{a}$	**M1**	Integrates to the form $k\arcsin\frac{x}{a}$, or substitutes $x = a\sin u$ and integrates. Condone a wrong constant k.
	$= \frac{2}{3a\lambda}\left(\frac{\pi}{2} + \frac{\pi}{6}\right)$	**dM1**	Substitutes both limits, or the corresponding limits in u, into their integral and subtracts.
	$r_m = \frac{4\pi}{9a\lambda}$	**A1**	oe
		[4]	
12(e)	$0.2395 \le a < 0.2405,\ 0.78 \le \lambda \le 0.82$ $\frac{4\pi}{9(0.2405)(0.82)} \le r_m \le \frac{4\pi}{9(0.2395)(0.78)}$	**M1**	Substitutes the greatest values of both a and λ, and the least of both, into their r_m of the form $\frac{k}{a\lambda}$. Condone 0.2404 for the upper bound of a.
	$7.08 \le r_m \le 7.47$ 7.2 lies in this range, so validity cannot be determined	**A1**	awrt 7.08 and 7.47, with a conclusion in words that the validity cannot be determined. A0 if the conclusion is missing.
		[2]	

Question 12 continued

Question	Scheme	Marks	Notes
12(f)	$v(0) = \frac{\sqrt{3}}{2}a\lambda = \frac{\sqrt{3}}{2}(0.2395)(0.78)$	**M1**	Uses v at $t = 0$ from their v, or $v^2 = \lambda^2\left(a^2 - \frac{a^2}{4}\right)$, with the lower bounds of both a and λ.
	$= 0.16\ (\mathrm{m\,s^{-1}})$	**A1**	
		[2]	
	Total for Question 12	**13**	